Grade 8 · Unit 1 · Cells and Body Systems — Flashcards

PA STEELS 3.1.6-8.A, B, C, G, H

GR 8 · UNIT 1 | STEELS 3.1.6-8.A | #1

Cell

The smallest unit that can be said to be alive. All living things are made of one or more cells.

Gr 8 · Unit 1 · #1 · 3.1.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.A | #2

How can you tell a living thing from a nonliving thing?

Living things are made of cells. Nonliving things, like a rock or a plastic toy, are not made of cells.

Gr 8 · Unit 1 · #2 · 3.1.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.A | #3

Unicellular

Made of only one cell. Bacteria are unicellular organisms.

Gr 8 · Unit 1 · #3 · 3.1.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.A #4

Multicellular

Made of many cells, often of different types and numbers. Humans, trees, and dogs are multicellular.

Gr 8 · Unit 1 · #4 · 3.1.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.B #5

How do the parts of a cell work together?

A cell works as a whole system. Each part has a special job, and all the jobs together keep the cell alive.

Gr 8 · Unit 1 · #5 · 3.1.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.B #6

Cell membrane

The thin boundary around every cell. It controls what enters and leaves the cell.

Gr 8 · Unit 1 · #6 · 3.1.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.B #7

Cell wall

A stiff layer outside the cell membrane of plant cells. It supports and protects the cell, helping plants stand up.

Gr 8 · Unit 1 · #7 · 3.1.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.B #8

Cell membrane vs. cell wall

All cells have a membrane that controls what goes in and out. Plant cells also have a stiff wall outside it for support.

Gr 8 · Unit 1 · #8 · 3.1.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.B #9

Nucleus

The part of the cell that holds its genetic information and controls the cell's activities.

Gr 8 · Unit 1 · #9 · 3.1.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.B #10

Chloroplast

The part of a plant cell that uses light energy to make food for the cell. Animal cells do not have chloroplasts.

Gr 8 · Unit 1 · #10 · 3.1.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.B #11

Mitochondria

Cell parts that release energy from food so the whole cell can do its work. Both plant and animal cells have them.

Gr 8 · Unit 1 · #11 · 3.1.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.C | #12

Tissue

A group of similar cells that work together to do a job. Example: muscle tissue.

Gr 8 · Unit 1 · #12 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.C | #13

Organ

A structure made of different tissues working together for a special body function. Examples: heart, lungs, stomach.

Gr 8 · Unit 1 · #13 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.C | #14

How is the body organized from cells?

Cells form tissues, tissues form organs, and organs work together in body systems.

Gr 8 · Unit 1 · #14 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.C | #15

Subsystem

A smaller system that is part of a bigger one. Each body system is a subsystem of the whole body.

Gr 8 · Unit 1 · #15 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.C #16

Circulatory system

The heart and blood vessels. It moves blood that carries oxygen and nutrients to all body cells.

Gr 8 · Unit 1 · #16 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.C #17

Respiratory system

The lungs and airways. It takes in oxygen and gets rid of carbon dioxide.

Gr 8 · Unit 1 · #17 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.C #18

Digestive system

Organs like the stomach and intestines. It breaks food down so the body can absorb nutrients.

Gr 8 · Unit 1 · #18 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.C #19

Excretory system

Organs like the kidneys and bladder. It removes wastes from the body.

Gr 8 · Unit 1 · #19 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.C | #20

Muscular system

The muscles of the body. They contract to move the body and to pump blood (heart muscle).

Gr 8 · Unit 1 · #20 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.C | #21

Nervous system

The brain, spinal cord, and nerves. It sends and processes signals that control the body.

Gr 8 · Unit 1 · #21 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.C | #22

How do body systems work together?

Systems depend on each other. During exercise, muscles need more oxygen, so breathing and heart rate both speed up.

Gr 8 · Unit 1 · #22 · 3.1.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.G | #23

What happens to food inside an organism?

Through chemical reactions, food molecules are broken apart and put back together to form new molecules.

Gr 8 · Unit 1 · #23 · 3.1.6-8.G

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.G #24

How does food help an organism grow?

Food is rearranged into new molecules that become part of the body, building new cells and tissues.

Gr 8 · Unit 1 · #24 · 3.1.6-8.G

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.G #25

How do organisms get energy from food?

When food molecules are broken apart in chemical reactions, stored energy is released for the organism to use.

Gr 8 · Unit 1 · #25 · 3.1.6-8.G

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.G #26

Cellular respiration

Chemical reactions in plant and animal cells where carbon-containing food molecules react with oxygen, releasing stored energy.

Gr 8 · Unit 1 · #26 · 3.1.6-8.G

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.G #27

What does cellular respiration produce?

Carbon dioxide and other materials, plus released energy the organism can use.

Gr 8 · Unit 1 · #27 · 3.1.6-8.G

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.G #28

Do plants carry out cellular respiration?

Yes. Plants and animals both use cellular respiration to release energy stored in food.

Gr 8 · Unit 1 · #28 · 3.1.6-8.G

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.H #29

Stimulus

Something in the environment that causes a response, like light, sound, heat, or a smell. Plural: stimuli.

Gr 8 · Unit 1 · #29 · 3.1.6-8.H

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.H #30

Sensory receptor

A part of a sense organ that detects a stimulus and sends a signal along nerve cells to the brain.

Gr 8 · Unit 1 · #30 · 3.1.6-8.H

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 STEELS 3.1.6-8.H #31

What kinds of inputs do sense receptors detect?

Electromagnetic (light for eyes), mechanical (touch, sound), and chemical (smell, taste). Each receptor responds to its own input.

Gr 8 · Unit 1 · #31 · 3.1.6-8.H

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.H | #32

Nerve cell

A cell that carries signals through the body, such as from sense receptors to the brain.

Gr 8 · Unit 1 · #32 · 3.1.6-8.H

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.H | #33

What does the brain do with signals from the senses?

It processes them. The result can be an immediate behavior (covering your ears at a loud noise) or a stored memory.

Gr 8 · Unit 1 · #33 · 3.1.6-8.H

✂ cut on dashes · fold on dots

GR 8 · UNIT 1 | STEELS 3.1.6-8.H | #34

Memory

Information the brain stores from sense signals so it can be used later, like remembering the smell of smoke means danger.

Gr 8 · Unit 1 · #34 · 3.1.6-8.H

✂ cut on dashes · fold on dots

Grade 8 · Unit 2 · Reproduction and Heredity — Flashcards

PA STEELS 3.1.6-8.D, E, M, N, R

GR 8 · UNIT 2 | STEELS 3.1.6-8.D | #1

Reproduction

The process by which organisms make offspring (new organisms of the same kind).

Gr 8 · Unit 2 · #1 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 | STEELS 3.1.6-8.D | #2

Offspring

The young, or new organisms, produced by a parent or parents.

Gr 8 · Unit 2 · #2 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 | STEELS 3.1.6-8.D | #3

Characteristic animal behavior

A typical behavior of a species that raises its odds of reproducing, like nest building or herding to protect young.

Gr 8 · Unit 2 · #3 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.D #4

How do some animals attract mates?

With behaviors and features like calls or songs (vocalization) and colorful feathers (plumage).

Gr 8 · Unit 2 · #4 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.D #5

How do some animals protect their young?

Building nests keeps young warm, and herding together protects young from predators. More young survive.

Gr 8 · Unit 2 · #5 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.D #6

Specialized plant structure

A plant part that helps it reproduce, like bright flowers, sweet nectar, strong odors, or hard nut shells.

Gr 8 · Unit 2 · #6 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.D #7

Pollen

Tiny grains made by flowers. For many plants to make seeds, pollen must be carried to a flower, often on another plant.

Gr 8 · Unit 2 · #7 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.D #8

Nectar

A sweet liquid in flowers that attracts insects and other animals, which carry pollen as they feed.

Gr 8 · Unit 2 · #8 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.D #9

How do animals help plants reproduce?

They carry pollen between flowers and move seeds. Squirrels bury nuts, and some grow into new plants.

Gr 8 · Unit 2 · #9 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.D #10

Seed germination

When a seed starts to sprout and grow into a new plant.

Gr 8 · Unit 2 · #10 · 3.1.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.E #11

Genetic factor

Inherited information (genes) from parents that affects growth. Example: large-breed cattle grow bigger than small breeds.

Gr 8 · Unit 2 · #11 · 3.1.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.E #12

Environmental factor

A local condition that affects growth, such as the amount of food, light, space, or water.

Gr 8 · Unit 2 · #12 · 3.1.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.E #13

How can the environment change how much an organism grows?

Drought slows plant growth and fertilizer speeds it up. Fish grow larger in big ponds than in small ponds.

Gr 8 · Unit 2 · #13 · 3.1.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.E #14

Why might two plants in the same garden grow differently?

They may have different genes (different seed varieties). Growth depends on both genetic factors and local conditions.

Gr 8 · Unit 2 · #14 · 3.1.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.M #15

Trait

A characteristic of an organism, such as eye color, height, or flower color.

Gr 8 · Unit 2 · #15 · 3.1.6-8.M

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.M #16

Chromosome

A structure in a cell that contains genes. Chromosomes come in pairs.

Gr 8 · Unit 2 · #16 · 3.1.6-8.M

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.M #17

Gene

A section of a chromosome that mainly controls the making of a specific protein, which affects a trait.

Gr 8 · Unit 2 · #17 · 3.1.6-8.M

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.M #18

Protein

A molecule a cell makes using instructions from a gene. Proteins affect an organism's structures, functions, and traits.

Gr 8 · Unit 2 · #18 · 3.1.6-8.M

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.M #19

How do genes affect traits?

A gene controls the making of a protein, and that protein affects the organism's traits.

Gr 8 · Unit 2 · #19 · 3.1.6-8.M

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.M #20

Mutation

A change in the structure of a gene. It may change a protein and therefore change a trait. Mutations are rare.

Gr 8 · Unit 2 · #20 · 3.1.6-8.M

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.M #21

Are mutations good or bad for an organism?

It depends. A mutation can be beneficial (helpful), harmful, or neutral (no effect) to the organism.

Gr 8 · Unit 2 · #21 · 3.1.6-8.M

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.N #22

Asexual reproduction

Reproduction with one parent. The offspring get identical genetic information, so they are copies of the parent.

Gr 8 · Unit 2 · #22 · 3.1.6-8.N

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.N #23

Sexual reproduction

Reproduction with two parents. Each parent gives half of the offspring's genes, so offspring have genetic variation.

Gr 8 · Unit 2 · #23 · 3.1.6-8.N

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.N #24

Genetic variation

Differences in genes between individuals. It causes differences in inherited traits.

Gr 8 · Unit 2 · #24 · 3.1.6-8.N

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.N #25

Inherited trait

A trait passed from parent(s) to offspring through genes, like the shape of a pea or a dog's fur color.

Gr 8 · Unit 2 · #25 · 3.1.6-8.N

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.N #26

Allele

One version of a gene. You have two alleles of each gene, one from each parent. They may be the same or different.

Gr 8 · Unit 2 · #26 · 3.1.6-8.N

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.N #27

Why do offspring from two parents look different from them?

Each parent passes on a random half of its chromosomes and genes, so offspring get a new mix of alleles.

Gr 8 · Unit 2 · #27 · 3.1.6-8.N

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.N #28

Punnett square

A grid model that shows the possible allele combinations offspring can get from two parents.

Gr 8 · Unit 2 · #28 · 3.1.6-8.N

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.N #29

Asexual vs. sexual reproduction

Asexual: one parent, offspring identical to parent. Sexual: two parents, offspring have genetic variation.

Gr 8 · Unit 2 · #29 · 3.1.6-8.N

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.R #30

Artificial selection

When humans choose which organisms reproduce so that desired traits are passed on to offspring.

Gr 8 · Unit 2 · #30 · 3.1.6-8.R

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.R #31

Selective breeding

Breeding parents with desired traits on purpose. Example: breeding cows that give the most milk.

Gr 8 · Unit 2 · #31 · 3.1.6-8.R

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.R #32

Animal husbandry

Raising and breeding farm animals, often choosing parents with traits people want, like more wool or meat.

Gr 8 · Unit 2 · #32 · 3.1.6-8.R

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.R #33

Genetic modification

Using technology to directly change an organism's genes to give it a desired trait, like crops that resist insects.

Gr 8 · Unit 2 · #33 · 3.1.6-8.R

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.R #34

Gene therapy

A medical technology that changes or replaces genes in a person's cells to treat a disease.

Gr 8 · Unit 2 · #34 · 3.1.6-8.R

✂ cut on dashes · fold on dots

GR 8 · UNIT 2 STEELS 3.1.6-8.R #35

How do trait-changing technologies affect society?

They can bring more food and new medical treatments, but people also debate their safety, cost, and fairness.

Gr 8 · Unit 2 · #35 · 3.1.6-8.R

✂ cut on dashes · fold on dots

Grade 8 · Unit 3 · Natural Selection and Evolution — Flashcards

PA STEELS 3.1.6-8.O, P, Q, S, T

GR 8 · UNIT 3 STEELS 3.1.6-8.O #1

Fossil record

All fossils found, placed in order of when they lived. It shows the history of life on Earth.

Gr 8 · Unit 3 · #1 · 3.1.6-8.O

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.O #2

How do scientists put fossils in time order?

By the sedimentary rock layer where they are found and by radioactive dating.

Gr 8 · Unit 3 · #2 · 3.1.6-8.O

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.O #3

In undisturbed rock layers, where are the oldest fossils?

Usually in the lowest (deepest) layers. Younger fossils are in layers above them.

Gr 8 · Unit 3 · #3 · 3.1.6-8.O

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.O #4

Radioactive dating

A method that uses radioactive elements in rocks to estimate how old the rocks and their fossils are.

Gr 8 · Unit 3 · #4 · 3.1.6-8.O

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.O #5

Extinction

When every member of a species has died out. The fossil record shows many extinct life forms.

Gr 8 · Unit 3 · #5 · 3.1.6-8.O

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.O #6

Diversity (of life)

The variety of different kinds of living things. The fossil record shows diversity has changed over Earth's history.

Gr 8 · Unit 3 · #6 · 3.1.6-8.O

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.O #7

What patterns does the fossil record show?

Life forms appeared, changed, and went extinct over time. Body structures generally became more complex in younger layers.

Gr 8 · Unit 3 · #7 · 3.1.6-8.O

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.O #8

Why can we use today's science to study ancient fossils?

Scientists assume natural laws work the same today as they did in the past.

Gr 8 · Unit 3 · #8 · 3.1.6-8.O

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.P #9

Anatomical structure

A body part of an organism, like a bone, wing, or fin.

Gr 8 · Unit 3 · #9 · 3.1.6-8.P

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.P #10

What do similar body structures suggest about organisms?

They suggest the organisms are related. Human arms, bat wings, and whale flippers have similar bones.

Gr 8 · Unit 3 · #10 · 3.1.6-8.P

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.P #11

Evolutionary relationship

How closely related organisms are through shared ancestors. More similar anatomy usually means a closer relationship.

Gr 8 · Unit 3 · #11 · 3.1.6-8.P

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 | STEELS 3.1.6-8.P | #12

Common ancestry

The idea that different species descended from the same ancestor long ago.

Gr 8 · Unit 3 · #12 · 3.1.6-8.P

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 | STEELS 3.1.6-8.P | #13

How do fossils help show evolutionary history?

Comparing anatomy of living organisms and fossils lets scientists infer lines of descent over time.

Gr 8 · Unit 3 · #13 · 3.1.6-8.P

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 | STEELS 3.1.6-8.Q | #14

Embryo

An organism in its earliest stage of development, before it is fully formed.

Gr 8 · Unit 3 · #14 · 3.1.6-8.Q

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 | STEELS 3.1.6-8.Q | #15

Embryological development

The changes an embryo goes through as it grows into a fully formed organism.

Gr 8 · Unit 3 · #15 · 3.1.6-8.Q

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.Q #16

What can comparing embryos show?

Embryos of different species can look alike early on, showing relationships not seen in the adults.

Gr 8 · Unit 3 · #16 · 3.1.6-8.Q

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.S #17

Natural selection

Individuals with traits that fit their environment survive and reproduce more, so those traits become common.

Gr 8 · Unit 3 · #17 · 3.1.6-8.S

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.S #18

Why do some individuals survive and reproduce more than others?

Genetic variation means some have traits that help in their environment. Example: green beetles hide well on leaves.

Gr 8 · Unit 3 · #18 · 3.1.6-8.S

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.S #19

What does natural selection do to traits in a population?

Helpful traits become predominant (more common) and less helpful traits are suppressed (become rare).

Gr 8 · Unit 3 · #19 · 3.1.6-8.S

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.S #20

How can probability describe natural selection?

If 3 of 4 dark moths survive on dark bark but only 1 of 4 light moths do, dark moths have a higher chance.

Gr 8 · Unit 3 · #20 · 3.1.6-8.S

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.T #21

Adaptation

A trait that helps an organism survive and reproduce in its environment. Natural selection over generations builds adaptations.

Gr 8 · Unit 3 · #21 · 3.1.6-8.T

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.T #22

Generation

One step in a family line: parents are one generation, and their offspring are the next.

Gr 8 · Unit 3 · #22 · 3.1.6-8.T

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.T #23

How do species change when the environment changes?

Over many generations, traits that help in the new environment become more common; others become less common.

Gr 8 · Unit 3 · #23 · 3.1.6-8.T

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.T #24

Distribution of traits

How common each trait is in a population. Natural selection changes this distribution over time.

Gr 8 · Unit 3 · #24 · 3.1.6-8.T

✂ cut on dashes · fold on dots

GR 8 · UNIT 3 STEELS 3.1.6-8.T #25

How can math show a trait changing in a population?

Track percentages over generations. Example: dark moths go from 10% to 40% to 70% of the population.

Gr 8 · Unit 3 · #25 · 3.1.6-8.T

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #1

Substance

A specific kind of matter, made from certain types of atoms. Water, salt, and oxygen are substances.

Gr 8 · Unit 4 · #1 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #2

Molecule

Two or more atoms joined together. Molecules range in size from two atoms to thousands of atoms.

Gr 8 · Unit 4 · #2 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #3

How do atoms make different substances?

Different types of atoms combine in different ways. Changing which atoms or how they join makes a different substance.

Gr 8 · Unit 4 · #3 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #4

Simple molecule

A molecule made of just a few atoms. Ammonia has 1 nitrogen and 3 hydrogen atoms.

Gr 8 · Unit 4 · #4 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #5

Extended structure

A solid made of a pattern of atoms (subunits) that repeats over and over. Examples: table salt and diamond.

Gr 8 · Unit 4 · #5 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #6

Crystal

A solid whose atoms or molecules are arranged in a repeating pattern. Salt and diamond are crystals.

Gr 8 · Unit 4 · #6 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #7

What is diamond made of?

Only carbon atoms, joined in a pattern that repeats in every direction.

Gr 8 · Unit 4 · #7 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #8

What is sodium chloride (table salt) made of?

Sodium and chlorine, in a repeating pattern with one sodium for every one chlorine.

Gr 8 · Unit 4 · #8 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.A #9

Ball-and-stick model

A 3D model where balls stand for atoms and sticks show how they are joined. Different colors mean different atoms.

Gr 8 · Unit 4 · #9 · 3.2.6-8.A

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #10

Pure substance

Matter made of only one kind of substance, like water, carbon dioxide, or helium.

Gr 8 · Unit 4 · #10 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #11

Particle

A tiny piece of matter, such as a molecule or an atom, that is always moving.

Gr 8 · Unit 4 · #11 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #12

Inert atom

An atom that does not easily combine with other atoms, so it moves around alone. Helium is an example.

Gr 8 · Unit 4 · #12 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #13

Kinetic energy

The energy of motion. Faster-moving particles have more kinetic energy.

Gr 8 · Unit 4 · #13 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #14

Thermal energy

The energy from the motion of the atoms or molecules inside a substance.

Gr 8 · Unit 4 · #14 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #15

Heat

In science, the thermal energy transferred from a warmer object to a cooler one because of a temperature difference.

Gr 8 · Unit 4 · #15 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #16

Temperature

A measure related to the average kinetic energy of a substance's particles. Faster particles mean a higher temperature.

Gr 8 · Unit 4 · #16 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #17

Is temperature the same as thermal energy?

No. A bathtub of warm water has more thermal energy than a cup at the same temperature, because it has more particles.

Gr 8 · Unit 4 · #17 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #18

How do particles move in a solid?

They are packed closely and vibrate in place. They do not change places with each other.

Gr 8 · Unit 4 · #18 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #19

How do particles move in a liquid?

They stay in contact with each other but slide past one another.

Gr 8 · Unit 4 · #19 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #20

How do particles move in a gas?

They are spread far apart and move freely, touching only when they happen to collide.

Gr 8 · Unit 4 · #20 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #21

What happens to particles when thermal energy is added?

They move faster (more kinetic energy) and temperature rises, until a change of state happens, like ice melting.

Gr 8 · Unit 4 · #21 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #22

What happens to particles when thermal energy is removed?

They slow down (less kinetic energy) and temperature drops, until a change of state happens, like water freezing.

Gr 8 · Unit 4 · #22 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #23

Change of state

When matter changes between solid, liquid, and gas, such as melting, freezing, or boiling.

Gr 8 · Unit 4 · #23 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.B #24

What can cause a change of state?

Changing the temperature or the pressure. Example: heating ice melts it, and squeezing a gas under high pressure can turn it into a liquid.

Gr 8 · Unit 4 · #24 · 3.2.6-8.B

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.C #25

Physical property

A trait you can observe or measure without changing the substance, like density, melting point, or boiling point.

Gr 8 · Unit 4 · #25 · 3.2.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.C #26

Chemical property

A trait that shows how a substance reacts to form new substances, like flammability (ability to burn).

Gr 8 · Unit 4 · #26 · 3.2.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.C #27

How can properties identify a substance?

Each pure substance has its own set of physical and chemical properties, like a fingerprint.

Gr 8 · Unit 4 · #27 · 3.2.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.C #28

Natural resource

A material from nature that people use, such as oil, plants, minerals, or water.

Gr 8 · Unit 4 · #28 · 3.2.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.C #29

Synthetic material

A material made by people through a chemical process using natural resources. Example: plastic made from oil.

Gr 8 · Unit 4 · #29 · 3.2.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.C #30

Chemical process

A process in which atoms of the starting substances regroup into new molecules with different properties.

Gr 8 · Unit 4 · #30 · 3.2.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 4 STEELS 3.2.6-8.C #31

How do synthetic materials affect society?

They give us new medicines, foods, and alternative fuels, but making them uses natural resources.

Gr 8 · Unit 4 · #31 · 3.2.6-8.C

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #1

Chemical reaction

A process where substances interact and their atoms regroup to form new substances with different properties.

Gr 8 · Unit 5 · #1 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #2

Reactant

A starting substance in a chemical reaction. Example: sugar is a reactant when sugar burns.

Gr 8 · Unit 5 · #2 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #3

How can you tell if a chemical reaction happened?

Compare properties before and after. If the new substance has different properties, a reaction occurred.

Gr 8 · Unit 5 · #3 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #4

Melting point

The temperature at which a solid turns into a liquid. Each pure substance has its own melting point.

Gr 8 · Unit 5 · #4 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #5

Boiling point

The temperature at which a liquid boils and turns into a gas. Water boils at 100 °C at sea level.

Gr 8 · Unit 5 · #5 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #6

Solubility

How well a substance dissolves in another, such as in water. Sugar is very soluble in water; sand is not.

Gr 8 · Unit 5 · #6 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #7

Flammability

How easily a substance burns. Paper and wood are flammable; sand is not.

Gr 8 · Unit 5 · #7 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #8

Odor

The smell of a substance. A new odor after substances mix can be evidence of a chemical reaction.

Gr 8 · Unit 5 · #8 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.D #9

How can density show that a reaction happened?

Each substance has its own density. If the density after mixing differs from the starting substances, a new substance may have formed.

Gr 8 · Unit 5 · #9 · 3.2.6-8.D

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.E #10

What happens to atoms in a chemical reaction?

The atoms in the reactants regroup into different molecules. No atoms are created or destroyed.

Gr 8 · Unit 5 · #10 · 3.2.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.E #11

Does the number of atoms change in a chemical reaction?

No. The total number of each type of atom is the same before and after the reaction.

Gr 8 · Unit 5 · #11 · 3.2.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.E #12

Law of conservation of matter

In a chemical reaction, matter is not created or destroyed. Atoms are conserved, so mass stays the same.

Gr 8 · Unit 5 · #12 · 3.2.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.E #13

Why does mass stay the same in a chemical reaction?

Because the same atoms are still there, just rearranged. Same atoms means same mass.

Gr 8 · Unit 5 · #13 · 3.2.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.E #14

If a gas escapes, why might a reaction seem to lose mass?

The gas atoms leave the container but are not destroyed. In a sealed container, the mass stays the same.

Gr 8 · Unit 5 · #14 · 3.2.6-8.E

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #15

Do all chemical reactions give off energy?

No. Some chemical reactions release energy, and others store (absorb) energy.

Gr 8 · Unit 5 · #15 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #16

What happens when a chemical process releases thermal energy?

The surroundings warm up. Dissolving calcium chloride in water makes the water feel hot.

Gr 8 · Unit 5 · #16 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #17

What happens when a chemical process absorbs thermal energy?

The surroundings cool down. Dissolving ammonium chloride in water makes the water feel cold.

Gr 8 · Unit 5 · #17 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #18

Concentration

How much of a substance is mixed into a certain amount of another, like grams of salt per cup of water.

Gr 8 · Unit 5 · #18 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #19

How can you change how hot or cold a chemical device gets?

Change the type of substance or its concentration. More substance can make a bigger temperature change.

Gr 8 · Unit 5 · #19 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #20

What do you measure when testing a hot or cold pack?

The amount of substance, the time, and the temperature reached.

Gr 8 · Unit 5 · #20 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #21

Why test and modify a design?

Test results show what works and what doesn't, so you can change the design to improve it.

Gr 8 · Unit 5 · #21 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #22

Redesign

Changing a design based on test results. You can combine the best features from different designs.

Gr 8 · Unit 5 · #22 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #23

Iterative process

Repeating the cycle of testing and modifying a design again and again to keep improving it.

Gr 8 · Unit 5 · #23 · 3.2.6-8.F

✂ cut on dashes · fold on dots

GR 8 · UNIT 5 STEELS 3.2.6-8.F #24

Optimal solution

The best design possible after many rounds of testing and improving.

Gr 8 · Unit 5 · #24 · 3.2.6-8.F

✂ cut on dashes · fold on dots